Part 4: Prism Dispersion and the Dark Side of the Moon

The cover of Pink Floyd's 1973 album *The Dark Side of the Moon* depicts a beam of white light entering a prism and emerging as a rainbow. In this assignment you will determine whether this depiction is physically consistent with the laws of refraction.

To do this you will construct a simple ray-tracing model that follows rays of light through a prism and calculates how they refract as a function of wavelength. Assume the prism is a 2-D equilateral triangle. White light enters the prism from the left.

The refractive index of the prism material varies with wavelength according to

$$n(\lambda) = 1.50 + \frac{0.004}{\lambda^2}$$

where λ is measured in microns. Your goal is to trace the path of several rays of different wavelengths through the prism and determine the direction in which they exit.

Wavelengths to Simulate

Use the following representative wavelengths:

Color	Wavelength (μm)
Violet	0.40
Blue	0.45
Cyan	0.50
Green	0.55
Yellow	0.58
Orange	0.62
Red	0.70

Tasks

Using your Jupyter notebook:

1. Implement a model that traces rays of light through the prism using Snell's law.
2. Calculate the path of each wavelength through the prism and determine the direction of the exiting ray.
3. Create a figure showing the paths of the rays through the prism
4. Answer the main question first based only on the wavelength dependence of $n(\lambda)$, and then again based on the ray-tracing figure you generated.

All figures generated by your notebook should be saved in the `figures` directory.

Model Verification

Before using your model to analyze the prism, you must demonstrate that the code behaves correctly. To do so, design and implement at least one verification experiment that confirms your code produces physically reasonable behavior. Include a short section in your notebook titled Model Verification that clearly describes the test you performed and explains why it demonstrates that the model is working properly.

AI Usage

Include a short section describing whether and how you used any AI tools to assist with coding or debugging. Include this as a short section at the end of your notebook.

Part 5: Submitting the Assignment

Homework will be submitted by providing a link to your GitHub repository on Canvas.

1. Make sure your repository contains the required structure:
```
sioc251_radiative_transfer
|
|-- README.md
|-- environment.yml
|
|-- HW0_prism_dispersion
    |
    |-- model.ipynb
    |
    |-- figures
        |-- one_or_more_figures.png
```
2. Open your notebook and select **Run All Cells**. This ensures that all figures and outputs are stored in the notebook.
3. Save the notebook after running all cells.
4. Upload your updated files to GitHub, either through the GitHub website or by using Git at the command line from the
`sioc251_radiative_transfer`
directory via:
```
git add .
git commit -m "Final commit for HW0"
git push -u origin main
```
5. Copy the URL of your GitHub repository and submit it on Canvas.

Your repository should allow someone else to view your notebook, figures, and written answers directly on GitHub without needing to run your code. Make sure your repository is set to **public** so that it can be viewed for grading.